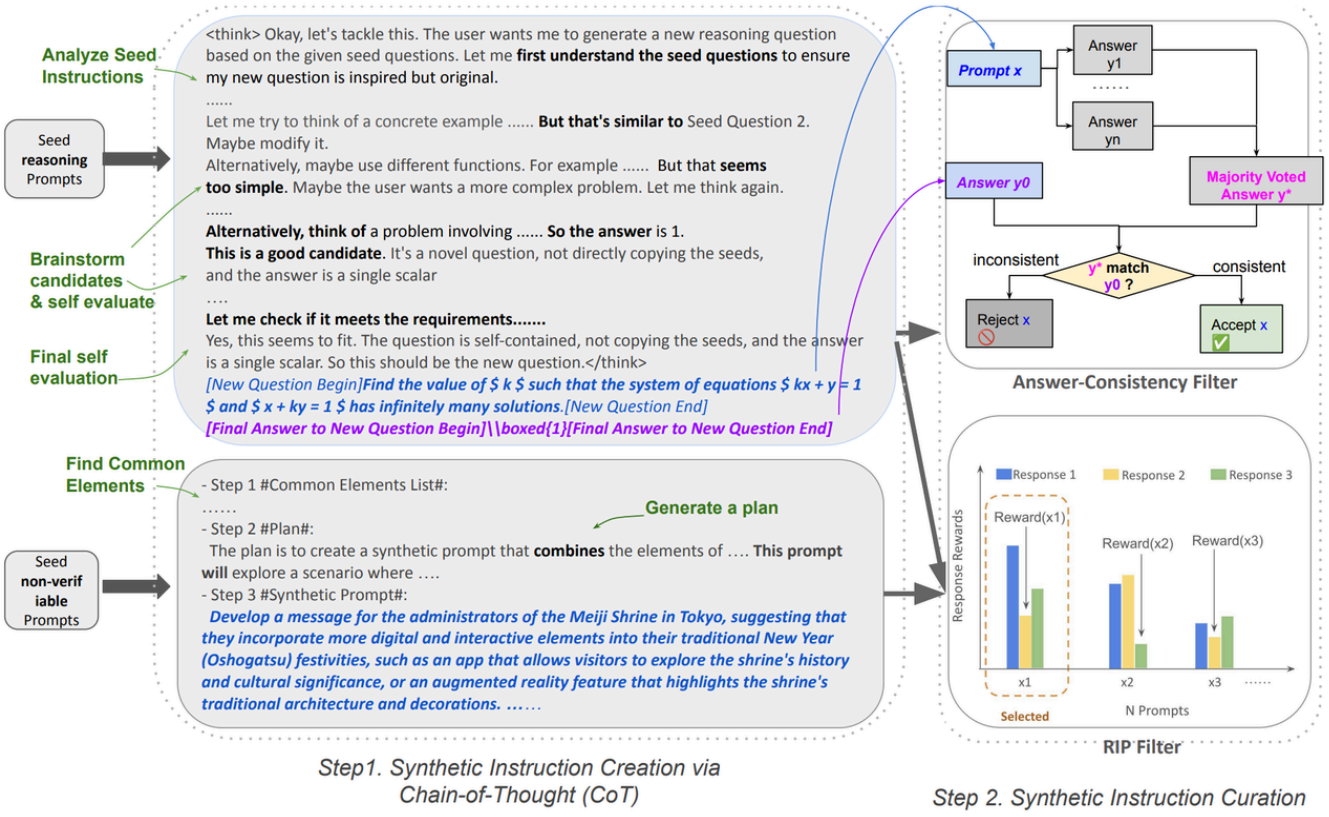
COT-SELF-INSTRUCT: BUILDING HIGH-QUALITY SYNTHETIC DATA FOR REASONING AND NON-REASONING TASKS

> #Dataset



- What problems does this framework solve?
  - The training of current large language models (LLMs) relies heavily on high-quality data, but manually acquiring such data is costly and prone to bias or errors.
  - For very complicated inference tasks, LLM-generated data faces the challenges of quality, especially for open-answer questions.
- What does this framework do?
  - Sample a set of seed data.
  - Data generation:
    - For verifiable data (math problems, e.g.):
      - The model is asked to analyze the seed data carefully by writing out the CoT, such as areas and complexity and finally produce a question
      - During the thinking process, the model is also asked to come up with an answer, which will significantly lower the difficulty of generating high quality QA pairs.
    - For open-answering data:
      - The model is only asked to analyze the seed data with the following steps:
        1. Identify common elements within the given seed instructions:

2. Develop a generation plan based on these elements

and finally produce a prompt **without answering it**.

- Data clean
  - For verifiable data: Use **major-voting** to generate a set of data independently. Discard the question if the voted answer is not equal to the answer generated in the data generation stage. (*Here we assume that the original data generated is correct, but is it always the case?*)
  - For non-verifiable data: For each multishot prompt, generate a set of candidates prompt (like paraphrase), and filter with the R.I.P mechanism and the keep the question with the highest R.I.P score among each of the set of candidates.
- Finetuning:
  - After acquiring high-quality synthetic data, the paper employs these datasets to perform reinforcement learning fine-tuning on the model: the GRPO algorithm is used for reasoning tasks, while the DPO algorithm is applied offline or online for general command tasks.

- Experiment Results
  - Comprehensive Outperformance of Existing Datasets: Models trained on data generated by CoT-Self-Instruct consistently outperformed counterparts trained on standard Self-Instruct, public datasets (e.g., s1K, OpenMathReasoning), and human-curated datasets (WildChat) across both reasoning benchmarks (MATH500, AMC23) and general benchmarks (AlpacaEval 2.0, Arena-Hard).
  - Quality over quantity: Experiments demonstrate that "a small amount of rigorously filtered data" significantly outperforms “unfiltered massive datasets.”
  - Process over outcome: Whether generating reasoning questions or open-ended prompts, forcing models to output longer CoT reasoning processes yields substantially higher-quality data compared to data generated without CoT or using shosrt CoT.
- My Thought
  - During data generation of verifiable questions, the answer generated along with the question is regarded as the ground truth, but is it always the truth? If the ground truth is wrong, and the major-voting produces another wrong answer, then this example is a bad one and may cause degrade of the policy model.
  - One possible solution is to use another frozen large monitor model to double judge the answer.
- Work Cited
  - Paper
  - Chat History
  - NotebookLM